Week 10: Learning – with Latent Random Variables Overview, Resources and Notes

1 Overview

For the parameter estimation techniques of Weeks 8 and 9, we assumed that all random variables (RVs) in our model are observed. However, it is often the case that not all RVs in a model are observed, and this is therefore a restricting assumption. For this week, we look at how to estimate parameters when some RV observations are missing.

Observations of RVs can be missing for several reasons, and in some cases, we need to model the mechanism that "decides" whether a certain RV is observed or not, and we have to include this mechanism in our model. However, we only focus on the simple case where some RVs are always observed (denoted by $\mathcal{D}$) and the other RVs are never observed (also called *hidden* or *latent* RVs; denoted by $\mathcal{H}$).

Just as for maximum-likelihood (ML) estimation, we might want to find values for the parameters, $\mathbf{\Theta}$, that maximise the *data likelihood*, $p(\mathcal{D}|\mathbf{\Theta})$. When there are latent RVs present in our model, this usually impossible to do in a single step. A feasible way to maximise the data likelihood[1] , $p(\mathcal{D}|\mathbf{\Theta})$, is to follow an iterative approach, where each iteration consists of two steps – this is the *expectation-maximisation (EM) algorithm*[2]. Similarly to ML estimation of Week 8, the EM algorithm produces point estimates (i.e., a set of values) for the parameters. We focus on (this version of) the EM algorithm for Week 10.

If we would instead like to find value for the parameters, $\mathbf{\Theta}$, that maximise the *posterior parameter distribution*, $p(\mathbf{\Theta}|\mathcal{D}) \propto p(\mathcal{D}|\mathbf{\Theta})p(\mathbf{\Theta})$, as used in maximum a posteriori (MAP) estimation, we can easily adapt the EM algorithm to maximise this objective. It essentially amounts to adding a regularisation term to the optimisation objective (just as we did for MAP estimation compared to ML estimation). Similarly to MAP estimation of Week 9, this version of the EM algorithm also produces point estimates for the parameters.

If we would rather like to follow the full Bayesian approach to estimate the parameters, it is easy to do in principle: consider all uncertain quantities ($\mathbf{\Theta}$, $\mathcal{D}$ and $\mathcal{H}$) to be RVs, construct a model for all these RVs, insert the evidence ($\mathcal{D}$) into the model, and perform inference to determine the posterior distribution over the parameters, $p(\mathbf{\Theta}|\mathcal{D})$[3]. Similarly to the full Bayesian approach to parameter estimation of Week 9, this approach produces distributions for the parameters. In practice, the full Bayesian approach might be very expensive or intractable though, and we do not focus on it in this module.

2 Resources

Prescribed resources

Study the following resources before the lectures and practical on Thursday (see the notes further down to get an idea of what to focus on):

- **Bishop** Chapter 9
- **Barber** Sections 11.1 and 11.2

[1] When latent RVs are present in our model, the data likelihood usually contains local maxima, and an iterative maximisation technique might get stuck in a local optimum.

[2] One can also use gradient-based optimisation approaches, but we don't use them in this module.

[3] We would not extract the joint posterior distribution over all the parameters, of course, but rather the marginal distributions over single parameters.

Supplementary resources (optional)

I also recommend that you study the following resources, if you have access to it:

- **Koller videos** in the Coursera course Probabilistic Graphical Models 3: Learning, Week 4:
  - *Learning With Incomplete Data: Learning With Incomplete Data – Overview (21:34)*
  - *Learning With Incomplete Data: Expectation Maximization – Intro (16:17)*
  - *Learning With Incomplete Data: Analysis of EM Algorithm (11:34)*
  - *Learning With Incomplete Data: EM in Practice (11:17)*
  - *Learning With Incomplete Data: Latent Variables (22:00)*

If you would like to read further, the following resources might be helpful: **Koller&Friedman** Section 19.2; The **PGM Tutorial** Section 6.4; **Murphy** Sections 6.5.3 to 6.5.6; **Kochenderfer** Section 4.4; **Jordan** Chapter 11.

Comments on the resources

- **Bishop** Chapter 9 uses the EM algorithm to learn mixture models from data. However, the EM algorithm is much more widely applicable – we can apply it to a wide range of cases where we want to learn parameters from data but where there are latent RVs present. Nevertheless, the fact that one can visualise mixture models well makes the presentation of the EM algorithm easy to grasp, and **Bishop** formulates the EM algorithm in a way that is not specific to mixture models. Take note of the following specific aspects of this chapter.
  - Section 9.1 discusses the k-means clustering algorithm, which we do not cover in this module. However, it is easy to understand, and it sets the scene for the rest of the chapter (one can think of the EM algorithm as solving the same problem as that of k-means, but doing so in a more principled and reliable way). I recommend that you quickly read through this section.
  - Section 9.2 introduces mixtures of Gaussians and presents the EM algorithm for learning mixtures of Gaussians from data. We do not focus on this specific form of the EM algorithm in the module, but the EM algorithm makes intuitive sense in the mixture of Gaussians setting. I recommend that you study this section.
  - Section 9.3 views the EM algorithm as parameter learning in the presence of latent RVs, which is the perspective we use in this module. It also describes the EM algorithm in a general way. This section is important to study. You can skip Sections 9.3.3 and 9.3.4 though.
  - Section 9.4 derives the EM algorithm for the general case. It is quite abstract and technical, but it is very important that you understand this section well – you can only effectively apply the EM algorithm in a wide range of settings if you understand the formulation presented in this section well.
- **Barber** Sections 11.1 and 11.2 cover missing data and the EM algorithm. Take note of the following specific aspects.
  - Section 11.1: This section discusses hidden (latent) variables and missing data. It is important to understand.
  - Section 11.2: This section presents the EM algorithm, and overlaps with much of **Bishop** Chapter 9. However, **Barber** presents the EM algorithm in a less thorough and less general way than **Bishop**. For this reason, I recommend that you use **Bishop** as your primary resource for the EM algorithm. The discussion and characterisation of the EM algorithm in **Barber** is valuable though. I recommend that you carefull *read through* this section. You can skip Section 11.2.6.

3 Notes

In my opinion, understanding the *derivation* of the EM algorithm is key to understanding the EM algorithm and how to use it. For this reason, I want to go through the derivation of the EM algorithm, which will cover much of the same ground as **Bishop** Chapter 9[4]. However, I found some parts of **Bishop** Chapter 9 hard to understand, and I tried to explain those parts more clearly in this section. So, I hope you find it valuable to study both **Bishop** Chapter 9 and the notes in this section.

Without any further ado, let us jump straight into the derivation of the EM algorithm. The objective we want to maximise in the EM algorithm is the data likelihood, $p(\mathcal{D}|\boldsymbol{\Theta})$. This can be written in terms of both the observed and latent RVs as[5]

$$p(\mathcal{D}|\boldsymbol{\Theta}) = \sum_{\mathcal{H}} p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta}). \tag{1}$$

The joint distribution, $p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})$, is constructed by multiplying all the factors of the model together, and is therefore usually in a nice factorised form. If $p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})$ had been the maximisation objective, then finding the parameters that maximise this objective would have been easy, since it decomposes nicely into local (i.e. small) likelihood functions. However, our maximisation objective is $p(\mathcal{D}|\boldsymbol{\Theta})$, which is usually impossible to maximise in a single step. But why is this the case? The fly in the ointment[6] is the marginalisation over the latent RVs, $\mathcal{H}$, as seen in Equation 1, which couples the remaining RVs, $\mathcal{D}$, and destroys the nice factorised format of $p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})$ (think back to the effect that the marginalisations of variable elimination (VE) had for a large, densely connected model). This usually causes $p(\mathcal{D}|\boldsymbol{\Theta})$ to contain a few very large factors, which is impossible to work with.

In order to get past this issue and work with the nice factorised distribution $p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})$ instead of with the ugly, coupled distribution $p(\mathcal{D}|\boldsymbol{\Theta})$, we use the following (somewhat strange) manipulation: First, we use the chain rule to write the data likelihood as

$$p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta}) = p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})p(\mathcal{D}|\boldsymbol{\Theta}) \Rightarrow p(\mathcal{D}|\boldsymbol{\Theta}) = \frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})}. \tag{2}$$

Next, we slightly change the maximisation objective: Since the logarithmic function is strictly monotonic, the maximum for $p(\mathcal{D}|\boldsymbol{\Theta})$ occurs at the same set of parameters as for $\log p(\mathcal{D}|\boldsymbol{\Theta})$. The latter is more convenient to work with, so we'll use it from this point onwards as the maximisation objective. Now we define an arbitrary distribution over the latent RVs, $q(\mathcal{H})$. Since this is by definition a (valid) probability distribution, we know that

$$\sum_{\mathcal{H}} q(\mathcal{H}) = 1. \tag{3}$$

Let's now combine Equations 2 and 3 to write the maximisation objective as

$$\begin{aligned}
\log p(\mathcal{D}|\boldsymbol{\Theta}) &= \log p(\mathcal{D}|\boldsymbol{\Theta}) \left[\sum_{\mathcal{H}} q(\mathcal{H})\right] \qquad \text{[use Eq. 3]} \\
&= \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}|\boldsymbol{\Theta}) \qquad \text{[move } \log p(\mathcal{D}|\boldsymbol{\Theta}) \text{ into sum]} \\
&= \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})}\right] \qquad \text{[use Eq. 2]} \\
&= \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})} \cdot \frac{q(\mathcal{H})}{q(\mathcal{H})}\right] \qquad \left[\text{multiply with } \frac{q(\mathcal{H})}{q(\mathcal{H})} = 1\right] \\
&= \sum_{\mathcal{H}} q(\mathcal{H}) \left\{\log \left[\frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{q(\mathcal{H})}\right] + \log \left[\frac{q(\mathcal{H})}{p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})}\right]\right\} \qquad [\log(a \cdot b) = \log a + \log b] \\
&= \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{q(\mathcal{H})}\right] - \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta})}{q(\mathcal{H})}\right] \qquad \text{[rearrange]}
\end{aligned} \tag{4}$$

It should be relatively easy for you to follow each step, but the reason for these manipulations might not be clear – it might seem that we have just made things more complicated. To make sense of it, let's look at each term on

[4]In fact, I wrote this section as a summary of **Bishop** Chapter 9 in a previous year before I knew that the **Bishop** textbook was publicly available.

[5]If the latent RVs are continuous, then the sum will be replace by an integral, of course.

[6]Afrikaans: "die vark in die verhaal"

the last line of Equation 4 in more detail. When you inspect the first term, you might notice that it contains the nice factorised distribution $p(\mathcal{D}, \mathcal{H}|\mathbf{\Theta})$ instead of the ugly distribution $p(\mathcal{D}|\mathbf{\Theta})$, and it should therefore be easy to maximise – this is exactly what we are going to do. We call this first term the variational lower bound, and denote it by

$$\mathcal{L}(q, \mathbf{\Theta}) \triangleq \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{D}, \mathcal{H}|\mathbf{\Theta})}{q(\mathcal{H})} \right]. \tag{5}$$

You might recognise the second term of Equation 4 as the Kullback-Leibler divergence between distributions $p(\mathcal{H}|\mathcal{D}, \mathbf{\Theta})$ and $q(\mathcal{H})$; this is denoted by

$$D_{\text{KL}}(q \parallel p) \triangleq -\sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{H}|\mathcal{D}, \mathbf{\Theta})}{q(\mathcal{H})} \right]. \tag{6}$$

You can think of the Kullback-Leibler divergence as the "distance" between two distributions (i.e., how similar the distributions are)[7]. The Kullback-Leibler divergence is always non-negative, and it is only 0 when the two distributions ($p(\mathcal{H}|\mathcal{D}, \mathbf{\Theta})$ and $q(\mathcal{H})$ in this case) are exactly the same.

So, we can write our maximisation objective of Equation 4 in terms of the variational lower bound of Equation 5 and the Kullback-Leibler divergence of Equation 6 as

$$\log p(\mathcal{D}|\mathbf{\Theta}) = \mathcal{L}(q, \mathbf{\Theta}) + D_{\text{KL}}(q \parallel p). \tag{7}$$

It should now be clear to you why $\mathcal{L}(q, \mathbf{\Theta})$ is a lower bound to $\log p(\mathcal{D}|\mathbf{\Theta})$, since $D_{\text{KL}}(q \parallel p)$ is always non-negative. The idea of the EM algorithm is to iteratively perform two steps, where each step improves one aspect of Equation 7 at a time: In the first step (the E-step), the parameters, $\mathbf{\Theta}$, are held fixed and the Kullback-Leibler divergence, $D_{\text{KL}}(q \parallel p)$, is minimised by choosing the distribution $q(\mathcal{H})$; this has the effect of fitting a better variational lower bound, $\mathcal{L}(q, \mathbf{\Theta})$. In the second step (the M-step), the distribution $q(\mathcal{H})$ is held fixed, and the parameters, $\mathbf{\Theta}$, that maximise the variational lower bound, $\mathcal{L}(q, \mathbf{\Theta})$, are found. This iterative process is illustrated in Figure 1.

Let's now look at each step in more detail.

The expectation step (E-step): Each iteration of the EM algorithm starts with the latest choice of parameter values, $\mathbf{\Theta}^{\text{old}}$ (for the first iteration, these are the initial parameter choices; for all other iterations, these are the parameters determined in the M-step of the previous iteration). In the E-step, the parameters, $\mathbf{\Theta}^{\text{old}}$, are held fixed, and the distribution $q(\mathcal{H})$ is chosen so that the variational lower bound, $\mathcal{L}(q, \mathbf{\Theta})$, at the current parameters, $\mathbf{\Theta}^{\text{old}}$, is fitted to the maximisation objective, $\log p(\mathcal{D}|\mathbf{\Theta})$ (see Figure 1c). This is done by choosing $q(\mathcal{H})$ such that the Kullback-Leibler divergence term of Equation 7 is 0. When you look at Equation 6, you'll see that this will (only) happen when we choose

$$q(\mathcal{H}) = p\left(\mathcal{H}|\mathcal{D}, \mathbf{\Theta}^{\text{old}}\right). \tag{8}$$

This means that we should perform inference to determine the posterior belief over the latent RVs, $\mathcal{H}$, given the observed RVs, $\mathcal{D}$, and the current parameter values, $\mathbf{\Theta}^{\text{old}}$, and set the distribution $q(\mathcal{H})$ to this belief. In order to do this, we can apply any of the (marginal) inference algorithms we studied earlier in this module.

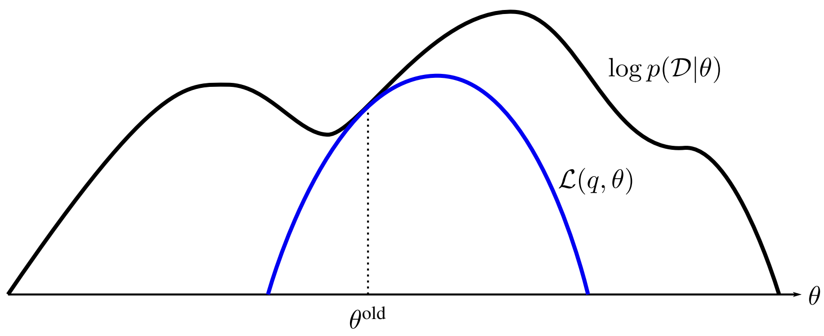
However, things are not quite as simple as they initially appear. To better understand what the inference of Equation 8 entails, let's look at the generic version of the model we use for parameter estimation as shown in Figure 2. In this figure, $\mathbf{d}^{[m]}$ refers to the observed RVs and $\mathbf{h}^{[m]}$ refers to the latent RVs associated with the mth data point. Since the parameters, $\mathbf{\Theta}^{\text{old}}$, are given (Eq. 8), it means that the model associated with one data point is independent of the model associated with another data point. Equation 8 therefore factorises as follows:

$$q(\mathcal{H}) = \prod_{m=1}^{M} q\left(\mathbf{h}^{[m]}\right) = \prod_{m=1}^{M} p\left(\mathbf{h}^{[m]}|\mathbf{d}^{[m]}, \mathbf{\Theta}^{\text{old}}\right). \tag{9}$$

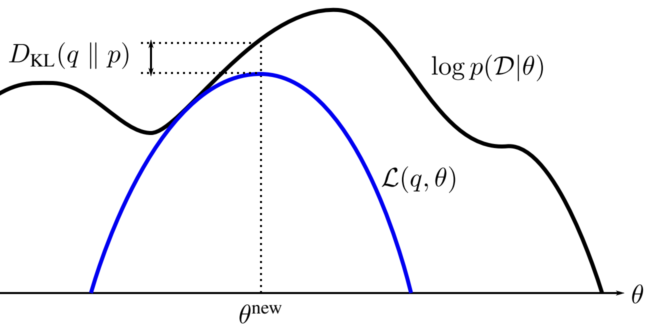
This factorisation is nice, since it means that inference is cheaper than it would otherwise be, but it also shows that we have to *repeat inference for each data point*.

A slightly different view of the E-step is that we perform a "soft completion" of the latent RVs in the E-step so that we can use the full data points (i.e., the observed RVs $\mathbf{d}^{[m]}$ and "completed" latent RVs $\mathbf{h}^{[m]}$) to estimate the parameters in the M-step.

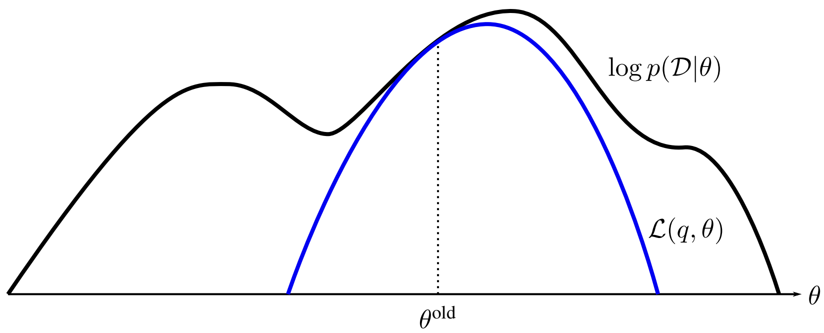
[7] It is not actually a distance, since it is not symmetric, but it is useful to think of it this way.



(a) During the E-step, the variational lower bound, $\mathcal{L}(q,\theta)$, is fitted to the maximisation objective, $\log p(\mathcal{D}|\theta)$, at the current parameter value, θ^{old}.



(b) During the M-step, the parameter θ^{new} that maximises the variational lower bound, $\mathcal{L}(q,\theta)$, is found. The Kullback-Leibler divergence, $D_{\text{KL}}(q \parallel p)$, at θ^{new} is now greater than 0.



(c) At the next E-step, the new parameter of the previous M-step, θ^{new}, becomes the old parameter, θ^{old}. The variational lower bound, $\mathcal{L}(q,\theta)$, is again fitted to the maximisation objective, $\log p(\mathcal{D}|\theta)$ at θ^{old} by setting the Kullback-Leibler divergence, $D_{\text{KL}}(q \parallel p)$, to 0; this is done by performing inference to determine $q(\mathcal{H})$.

Figure 1: Illustration of the EM algorithm for a single parameter, $\mathbf{\Theta} = \theta$

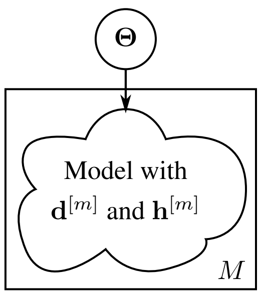


Figure 2: Generic description of the model we use in parameter estimation, drawn in plate notation

The maximisation step (M-step): In the M-step, we keep the choice of distribution $q(\mathcal{H})$ fixed to that we determined in the E-step; this fixes the "shape" of the variational lower bound, $\mathcal{L}(q, \boldsymbol{\Theta})$. The parameters are then updated such that the variational lower bound, $\mathcal{L}(q, \boldsymbol{\Theta})$, is maximised (see Figure 1b). To see how this is done, let's rewrite Equation 5 as

$$\begin{aligned}\mathcal{L}(q, \boldsymbol{\Theta}) &\triangleq \sum_{\mathcal{H}} q(\mathcal{H}) \log \left[\frac{p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})}{q(\mathcal{H})}\right] \\ &= \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta}) - \sum_{\mathcal{H}} q(\mathcal{H}) \log q(\mathcal{H}).\end{aligned} \tag{10}$$

The last term is independent of $\boldsymbol{\Theta}$, so it plays no role in the maximisation. We can therefore define the auxiliary function

$$\mathcal{Q}(\boldsymbol{\Theta}) \triangleq \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta}) \tag{11}$$

and determine updated values for the parameter by calculating

$$\boldsymbol{\Theta}^{\text{new}} = \arg\max_{\boldsymbol{\Theta}} \mathcal{Q}(\boldsymbol{\Theta}). \tag{12}$$

Exactly how the updated parameters are calculated depends on the specific model, and will have to be worked out specifically for each model. The manipulations are similar to that of ML and MAP estimation of Week 8 and 9.

EM algorithm summary: Given the development given so far, we can summarise the EM algorithm as shown in Algorithm 1.

Algorithm 1 The expectation-maximisation (EM) algorithm

Initialise $\boldsymbol{\Theta}^{\text{old}}$
while not converged **do**
  $q(\mathcal{H}) \leftarrow p\left(\mathcal{H}|\mathcal{D}, \boldsymbol{\Theta}^{\text{old}}\right)$ ▷ E-step
  $\boldsymbol{\Theta}^{\text{new}} \leftarrow \arg\max_{\boldsymbol{\Theta}} Q(\boldsymbol{\Theta})$, where $Q(\boldsymbol{\Theta}) = \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H}|\boldsymbol{\Theta})$ ▷ M-step
  $\boldsymbol{\Theta}^{\text{old}} \leftarrow \boldsymbol{\Theta}^{\text{new}}$
end while

Notes: Take note of the following aspects of the EM algorithm (see the resources for details):

- The EM algorithm is guaranteed to make progress with each iteration if it isn't already at a static point of the maximisation objective $\log p(\mathcal{D}|\boldsymbol{\Theta})$ (i.e., where $\log p(\mathcal{D}|\boldsymbol{\Theta})$ is "flat").
- The EM algorithm typically makes quick progress initially, but might be slow to converge.
- The EM algorithm can (and does) get stuck in local maxima. It is therefore important to initialise the parameters well.
- If you rather want to use the maximisation objective $\log p(\boldsymbol{\Theta}|\mathcal{D})$ instead of $\log p(\mathcal{D}|\boldsymbol{\Theta})$ (as you did for MAP vs. ML estimation in Week 10), it is equivalent to using the objective $\log p(\mathcal{D}|\boldsymbol{\Theta})p(\boldsymbol{\Theta}) = \log p(\mathcal{D}|\boldsymbol{\Theta}) + \log p(\boldsymbol{\Theta})$. You can implement this by grouping the new term, $\log p(\boldsymbol{\Theta})$, with the variational lower bound, $\mathcal{L}(q, \boldsymbol{\Theta})$, and maximising $\mathcal{L}(q, \boldsymbol{\Theta}) + \log p(\boldsymbol{\Theta})$ in the M-step. This will include the prior belief over the parameters in the parameter estimates (or, from a different perspective, add regularisation to the parameter estimation).

4 Example

Let's now apply the EM algorithm by working through an example. Consider the scnenario where data values are randomly drawn from two Gaussian distributions, but the means of the Gaussian distributions are unknown. This scenario could – for example – occur when an analogue circuit is used to implement a binary channel. This channel could represent bits (0 or 1) with two different voltage levels, but the value of these voltage levels might be unknown. The measurements are also noisy, and they therefore do not correspond exactly to one of the two voltage levels. This scenario is depicted in Figure 3.

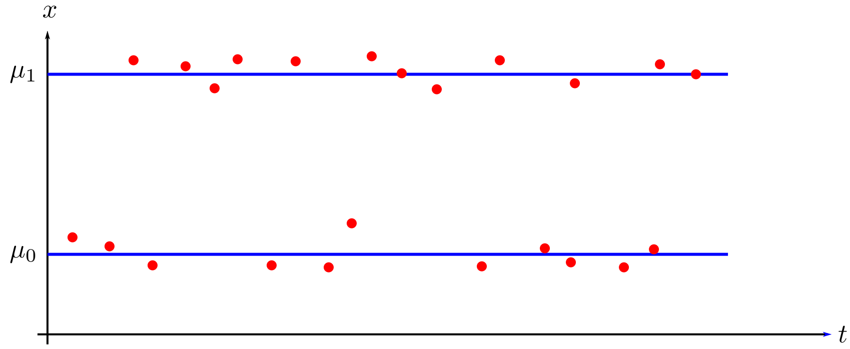


Figure 3: An illustration of the scenario used in the example. The blue lines represent the actual voltage levels corresponding to a 0 and 1 respectively, and the red dots represent the noisy measurements.

This scenario can be modelled with the Bayes network shown in Figure 4. It models how M observed values, $\mathcal{D} =$

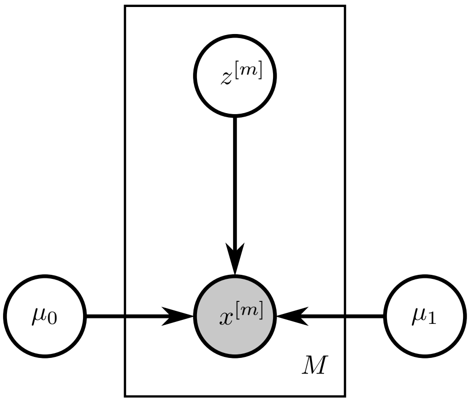


Figure 4: The model used in the example, drawn in plate notation

$\left(x^{[1]}, \ldots, x^{[M]}\right)$, are generated. The model contains M unobserved (latent) binary RVs, $\mathcal{H} = \left(z^{[1]}, \ldots, z^{[M]}\right)$, each having the domain $z^{[m]} \in \{0, 1\}$. The model also contains two (unknown) real parameters, μ_0 and μ_1. If $z^{[m]} = 0$, then the observed value $x^{[m]}$ is drawn from a Gaussian distribution with a mean of μ_0, or

$$p\left(x^{[m]} | z^{[m]} = 0, \mu_0, \mu_1\right) = \mathcal{N}\left(x^{[m]}; \mu_0, \sigma_v^2\right),$$

where σ_v^2 is a known variance. Conversely, if $z^{[m]} = 1$, then the observed value $x^{[m]}$ is drawn from a Gaussian distribution with a mean of μ_1, or

$$p\left(x^{[m]} | z^{[m]} = 1, \mu_0, \mu_1\right) = \mathcal{N}\left(x^{[m]}; \mu_1, \sigma_v^2\right).$$

The model does not have a higher prior preference for one Gaussian distribution; that is, $p\left(z^{[m]}\right)$ is uniform. The unknown parameters are given by $\boldsymbol{\Theta} = (\mu_0, \mu_1)$, and since the model contains both observed and latent RVs, we are going to use the EM algorithm to estimate the unknown parameters.

Let's first write the model in a format we can work with (i.e., as tables). The two tables for the factors corresponding to observation m in the plate of Figure 4 are given by

$z^{[m]}$	$p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right)$
0	$\mathcal{N}\left(x^{[m]}; \mu_0, \sigma_v^2\right)$
1	$\mathcal{N}\left(x^{[m]}; \mu_1, \sigma_v^2\right)$

$z^{[m]}$	$p\left(z^{[m]}\right)$
0	0.5
1	0.5

The full model is then given by the product of the factors, or

$$p(\mathcal{D}, \mathcal{H} \mid \boldsymbol{\Theta}) = \prod_{m=1}^{M} p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right) p\left(z^{[m]}\right). \tag{13}$$

With the model written in the format necessary for the EM algorithm, let's now work through the **E-step**. At the start of the E-step, the current parameter values are given by $\boldsymbol{\Theta}^{\text{old}} = \left(\mu_0^{\text{old}}, \mu_1^{\text{old}}\right)$, and we should determine $q(\mathcal{H}) = p\left(\mathcal{H} \mid \mathcal{D}, \boldsymbol{\Theta}^{\text{old}}\right)$ by performing inference. As shown in Equation 8, the distribution $q(\mathcal{H})$ factorises according to

$$q(\mathcal{H}) = \prod_{m=1}^{M} q\left(z^{[m]}\right) = \prod_{m=1}^{M} p\left(z^{[m]} \mid x^{[m]}, \mu_0^{\text{old}}, \mu_1^{\text{old}}\right). \tag{14}$$

We therefore have to perform inference over each data point. Since the model for each data point is very simple, we will perform inference by simply multiply the relevant factors together; the unnormalised posterior belief over each data point, $\tilde{q}\left(z^{[m]}\right)$, is therefore given by

$$p\left(z^{[m]} \mid x^{[m]}, \mu_0^{\text{old}}, \mu_1^{\text{old}}\right) \propto p\left(x^{[m]} \mid z^{[m]}, \mu_0^{\text{old}}, \mu_1^{\text{old}}\right) p\left(z^{[m]}\right) = \tilde{q}\left(z^{[m]}\right), \tag{15}$$

which – expressed as a table and using the definitions of the factors – is given by

$z^{[m]}$	$\tilde{q}\left(z^{[m]}\right)$
0	$0.5 \cdot \mathcal{N}\left(x^{[m]}; \mu_0^{\text{old}}, \sigma_v^2\right)$
1	$0.5 \cdot \mathcal{N}\left(x^{[m]}; \mu_1^{\text{old}}, \sigma_v^2\right)$

After normalising, the posterior belief over the latent RVs is given by

$z^{[m]}$	$q\left(z^{[m]}\right)$
0	$\gamma_0^{[m]} = \mathcal{N}\left(x^{[m]}; \mu_0^{\text{old}}, \sigma_v^2\right) / \left[\mathcal{N}\left(x^{[m]}; \mu_1^{\text{old}}, \sigma_v^2\right) + \mathcal{N}\left(x^{[m]}; \mu_0^{\text{old}}, \sigma_v^2\right)\right]$
1	$\gamma_1^{[m]} = \mathcal{N}\left(x^{[m]}; \mu_1^{\text{old}}, \sigma_v^2\right) / \left[\mathcal{N}\left(x^{[m]}; \mu_1^{\text{old}}, \sigma_v^2\right) + \mathcal{N}\left(x^{[m]}; \mu_0^{\text{old}}, \sigma_v^2\right)\right]$

The entries in this table, $\gamma_0^{[m]}$ and $\gamma_1^{[m]}$, are called the *responsibilities*[8], and can be interpreted as the probability that data point m has been generated by Gaussian component 0 or 1 respectively, given the current parameters, $\boldsymbol{\Theta}^{\text{old}}$. This is the end of the E-step, and these responsibilities will be used in the M-step.

For the **M-step**, let's first calculate the auxiliary function $\mathcal{Q}(\boldsymbol{\Theta})$:

$$\begin{aligned}
\mathcal{Q}(\boldsymbol{\Theta}) &= \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H} \mid \boldsymbol{\Theta}) \\
&= \sum_{z^{[1]}, \ldots, z^{[M]}} \left[\prod_{n=1}^{M} q\left(z^{[n]}\right)\right] \log \left[\prod_{m=1}^{M} p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right) p\left(z^{[m]}\right)\right] && \text{[Eqs. 13 and 14]} \\
&= \sum_{z^{[1]}, \ldots, z^{[M]}} \left[\prod_{n=1}^{M} q\left(z^{[n]}\right)\right] \sum_{m=1}^{M} \left[\log p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right) + \log p\left(z^{[m]}\right)\right] && \text{[log into product]} \\
&= \sum_{m=1}^{M} \left\{ \sum_{z^{[1]}, \ldots, z^{[M]}} \left[\prod_{n=1}^{M} q\left(z^{[n]}\right)\right] \cdot \left[\log p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right) + \log p\left(z^{[m]}\right)\right] \right\} && \text{[take sum out]} \\
&= \sum_{m=1}^{M} \left[\sum_{z^{[m]}} q\left(z^{[m]}\right) \log p\left(x^{[m]} \mid z^{[m]}, \mu_0, \mu_1\right)\right] + K.
\end{aligned} \tag{16}$$

[8] Note that all the quantities in this table ($x^{[m]}$, μ_0^{old}, μ_1^{old} and σ_v^2) are known at this point, so the responsibilities, $\gamma_0^{[m]}$ and $\gamma_1^{[m]}$, are constants. We only name them to shorten the expressions in the M-step.

To get from the second-to-last to the last line of Equation 16, we "push the (inner) sum into the product" for every $n \neq m$, which reduces to $\sum_{z^{[n]}} q\left(z^{[n]}\right) = 1$; the only time things do not reduce to 1 is when $n = m$. The term K represents all terms that are independent of the parameters, μ_0 and μ_1. By using the definition of $p\left(x^{[m]}|z^{[m]}, \mu_0, \mu_1\right)$ given earlier and applying the definition of the Gaussian distribution, as well as using $q\left(z^{[m]}\right)$ calculated in the E-step, we can continue our manipulation of Equation 16 as follows:

$$\begin{aligned}\mathcal{Q}(\boldsymbol{\Theta}) = & \sum_{m=1}^{M}\left[\gamma_0^{[m]}\left(-\frac{1}{2\sigma_v^2}\left(x^{[m]}-\mu_0\right)^2 - \log\sqrt{2\pi\sigma_v^2}\right)\right.\\ & \left. + \gamma_1^{[m]}\left(-\frac{1}{2\sigma_v^2}\left(x^{[m]}-\mu_1\right)^2 - \log\sqrt{2\pi\sigma_v^2}\right)\right] + K \\ = & -\frac{1}{2\sigma_v^2}\sum_{m=1}^{M}\left[\gamma_0^{[m]}\left(x^{[m]}-\mu_0\right)^2 + \gamma_1^{[m]}\left(x^{[m]}-\mu_1\right)^2\right] + K.\end{aligned} \tag{17}$$

To find the parameter value for μ_0 that maximises $\mathcal{Q}(\boldsymbol{\Theta})$, we take the partial derivative with respect to μ_0 and set it to 0:

$$\begin{aligned}&\frac{\partial}{\partial\mu_0}\mathcal{Q}(\boldsymbol{\Theta})\bigg|_{\mu_0=\mu_0^{\text{new}}} = -\frac{1}{2\sigma_v^2}\sum_{m=1}^{M}\left[-2\cdot\gamma_0^{[m]}\left(x^{[m]}-\mu_0^{\text{new}}\right)+0\right] = 0 \\ &\Rightarrow \sum_{m=1}^{M}\gamma_0^{[m]}x^{[m]} = \sum_{m=1}^{M}\gamma_0^{[m]}\mu_0^{\text{new}} \\ &\Rightarrow \mu_0^{\text{new}} = \left(\sum_{m=1}^{M}\gamma_0^{[m]}x^{[m]}\right)\bigg/\left(\sum_{m=1}^{M}\gamma_0^{[m]}\right).\end{aligned} \tag{18}$$

Similarly,

$$\frac{\partial}{\partial\mu_1}\mathcal{Q}(\boldsymbol{\Theta})\bigg|_{\mu_1=\mu_1^{\text{new}}} = 0 \quad \Rightarrow \mu_1^{\text{new}} = \left(\sum_{m=1}^{M}\gamma_1^{[m]}x^{[m]}\right)\bigg/\left(\sum_{m=1}^{M}\gamma_1^{[m]}\right). \tag{19}$$

Equations 18 and 19 specifies how to update the parameters, which completes the M-step.

We have now specified the calculations to be performed in the E-step and the M-step of the EM algorithm. By choosing initial values for the parameter estimates, and then repeatedly alternating between the E-step and the M-step until things converge, we can estimate the parameters μ_0 and μ_1.